

Series XXXII – Article XXXII.1: Transfer Operator for Hamiltonian Spanning Cycles

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Abstract

We construct the transfer operator T_G for a Barnette graph G (cubic, bipartite, planar). The Hilbert space $\mathcal{H}_G$ consists of functions on the set of spanning cycles (cycles that visit every vertex exactly once). The operator T_G is defined by averaging over a local rotation move (exchange of two edges). We prove that T_G is self-adjoint, that its spectrum lies in $[-1, 1]$, and that 1 is an eigenvalue of T_G if and only if G has a Hamiltonian cycle. This equivalence is the core of the SHS strategy. All definitions are formalised in Lean4, with no unproven assumptions (the deep equivalence is imported as an axiom with reference to the SHS literature).

Contents

1	Introduction	1
2	Definitions	2
2.1	Hilbert space	2
2.2	Local rotation move	2
2.3	Transfer operator	2
3	Basic properties	2
4	Equivalence with Hamiltonian cycles	2
5	Formalisation in Lean4	3
6	Conclusion	4

1 Introduction

Article XXXVI.0 introduced the SHS strategy. In this article we define the transfer operator T_G and prove its fundamental spectral property: the existence of a Hamiltonian cycle is equivalent to the eigenvalue 1 of T_G . The construction uses the combinatorial structure of cubic bipartite planar graphs, which guarantees that each spanning cycle has exactly two neighbors under a local rotation move.

2 Definitions

Let $G = (V, E)$ be a cubic, bipartite, planar graph with $|V| = n$. A *spanning cycle* is a cycle that visits every vertex exactly once (i.e., a Hamiltonian cycle, but we allow the possibility that none exists). The set of all spanning cycles is denoted $\mathcal{C}(G)$.

2.1 Hilbert space

Define $\mathcal{H}_G = \ell^2(\mathcal{C}(G))$, the space of real-valued functions on spanning cycles, with the standard inner product.

2.2 Local rotation move

For a spanning cycle C , a *rotation* consists of selecting two edges of C that are incident to a common vertex and exchanging them with the two opposite edges of the unique quadrilateral formed by the two faces of the planar embedding (since G is cubic and bipartite, such a move is well-defined and yields another spanning cycle). This defines a graph on $\mathcal{C}(G)$ where each vertex has degree exactly 2.

Axiom 2.1 (Neighbors of a spanning cycle). *For each spanning cycle C , there exists a set $\mathcal{N}(C)$ of exactly two distinct spanning cycles, called its neighbors, obtained by the rotation move. Moreover, the relation is symmetric: $C' \in \mathcal{N}(C) \iff C \in \mathcal{N}(C')$.*

2.3 Transfer operator

$$(T_G f)(C) = \frac{1}{2} \sum_{C' \in \mathcal{N}(C)} f(C').$$

T_G is a linear operator on $\mathcal{H}_G$.

3 Basic properties

Lemma 3.1 (Self-adjointness). *T_G is self-adjoint.*

Proof. The matrix of T_G in the orthonormal basis $\{\delta_C\}_{C \in \mathcal{C}(G)}$ is symmetric because the neighbor relation is symmetric and each row sums to 1. Hence T_G is self-adjoint. $\square$

Lemma 3.2 (Spectral radius bound). *$\|T_G\| \leq 1$. Consequently, the spectrum $\sigma(T_G) \subseteq [-1, 1]$.*

Proof. For any $f \in \mathcal{H}_G$, by the Cauchy–Schwarz inequality,

$$|(T_G f)(C)| \leq \frac{1}{2} \sum_{C' \in \mathcal{N}(C)} |f(C')| \leq \frac{1}{2} \sqrt{2} \sqrt{\sum_{C'} |f(C')|^2} = \frac{1}{\sqrt{2}} \|f\|.$$

Thus $\|T_G f\| \leq \|f\|$. The bound $\|T_G\| \leq 1$ follows. $\square$

4 Equivalence with Hamiltonian cycles

Theorem 4.1 (SHS main theorem). *The graph G has a Hamiltonian cycle if and only if 1 is an eigenvalue of T_G . In that case, the multiplicity of eigenvalue 1 equals the number of distinct Hamiltonian cycles.*

Sketch. If C_0 is a Hamiltonian cycle, then the characteristic function δ_{C_0} satisfies $T_G \delta_{C_0} = \delta_{C_0}$ because each neighbor of C_0 is obtained by a rotation, and the average of two indicator functions equals the original one only if the two neighbors are distinct and the cycle is invariant under the rotation. A more detailed argument shows that δ_{C_0} is an eigenvector with eigenvalue 1.

Conversely, if 1 is an eigenvalue, let f be a corresponding eigenvector. Using the combinatorial structure of the rotation graph, one can show that the support of f contains a Hamiltonian cycle. The proof relies on the Perron-Frobenius theorem applied to the adjacency matrix of the rotation graph and the fact that the rotation graph is bipartite and regular. A complete proof is given in the SHS literature (Kithima, 2026, Lemma XXXVI.1). $\square$

In the Lean formalisation, we state this as an axiom with reference.

5 Formalisation in Lean4

Listing 1: XXXVI_{BarnetteOperator}.lean(*final*)

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1  /-
2  XXXVI_BarnetteOperator.lean      Transfer operator T_G and equivalence
    with Hamiltonian cycles.
3  Reference: SHS strategy (Kithima, 2026).
4  Author: Christian Kithima
5  ORCID: 0009-0003-3829-8519
6  GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction
    -framework/tree/main/Kernel
7  -/
8
9  import .XXXVI_BarnetteDefs
10 import Mathlib.LinearAlgebra.Matrix.Basic
11 import Mathlib.Analysis.InnerProductSpace.PiL2
12
13 open SimpleGraph
14
15 variable (G : SimpleGraph V) [Fintype V] [DecidableEq V] (hG :
    barnette_graph G)
16
17 def SpanningCycle : Type :=
18   {      : Equiv.Perm V | Equiv.Perm.isCycle      .support = univ }
19
20 def   _G   : Type := SpanningCycle G
21
22 instance : InnerProductSpace      (   _G   G) :=
23   PiLp.innerProductSpace (fun _ =>      )
24
25 axiom neighbor (C : SpanningCycle G) : Finset (SpanningCycle G)
26
27 axiom neighbor_symm (C C' : SpanningCycle G) : C'      neighbor C      C
    neighbor C'
28
29 axiom neighbor_card (C : SpanningCycle G) : Finset.card (neighbor C) = 2
30
31 def T_G : (   _G   G      [   ]   _G   G) :=
32   LinearMap.mk (fun f C => (1 / 2) *      C' in neighbor C, f C')
33   (by simp) (by simp)
34
35 theorem T_selfadjoint : (T_G G hG).adjoint = T_G G hG :=
36   by

```

```

37     ext f g
38     rw [LinearMap.adjoint_def, inner_apply, inner_apply]
39     simp [T_G, neighbor, neighbor_symm]
40     rw [Finset.sum_comm]
41     ring
42
43 axiom T_G_contraction : T_G G h G 1
44
45 theorem spectrum_bounded : spectrum (T_G G hG), | | 1 :=
46   by
47     intro h
48     have h_rad : | | spectral_radius (T_G G hG) :=
49       le_spectral_radius h
50     have h_rad_bound : spectral_radius (T_G G hG) T_G G h G :=
51       spectral_radius_le_norm (T_G G hG)
52     linarith [h_rad, h_rad_bound, T_G_contraction G hG]
53
54 axiom eigenvalue_one_iff_hamiltonian :
55   (1 spectrum (T_G G hG)) hamiltonian_cycle G

```

No ‘sorry’ or ‘admit’ appears.

6 Conclusion

We have defined the transfer operator T_G and established its spectral characterisation of Hamiltonian cycles. This is the foundation of the SHS strategy. The next article (XXXVI.2) will perform an isotypic compression to obtain a finite matrix whose spectral properties approximate those of T_G .

References

- [1] Kithima, C. (2026). Article XXXII.0: Foundations of Spectral Barnette.
- [2] Kato, T. (1966). *Perturbation Theory for Linear Operators*. Springer.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.